BreastScan: An AI- Powered Mobile Application for Breast Cancer Prediction

EPICS Phase – II Report

Submitted by

SMRITI PRAJAPATI (23BCE11456)
HRITHEMA (23BCE11243)
ADITI RAJ (23BAI10178)
TANMAY AGNIHOTRI (23BAI10341)
GHAZIAH SHOEB (23BHI10034) AADITYA JAIN (23BCE10963)

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MADHYA PRADESH – 466114

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BONAFIDE CERTIFICATE

Certified that this project report titled **"BreastScan: An AI- Powered Mobile Application for Breast Cancer Prediction"** is the Bonafide work of **"Smriti Prajapati** (23BCE11456); **Hrithema** (23BCE11243); **Aditi Raj** (23BAI10178); **Tanmay Agnihotri** (23BAI10341); **Ghaziah Shoeb** (23BHI10034); **Aaditya Jain** (23BCE110963)**"** who carried out the project work under my supervision. Certified further that to the best of my knowledge, the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

PROJECT GUIDE
Dr. Praveen Lalwani
Assistant Professor
School of Computer Science and Engineering
VIT BHOPAL UNIVERSITY

REVIEWER 1

Dr. Buvaneswari P. R.
Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal University

REVIEWER 2

Dr. Ankit Pal
Assistant Professor
School of Applied Science and Language
VIT Bhopal University

The EPICS Phase 2 is held on ____

Declaration of Originality

We hereby declare that this report entitled **"BreastScan: An AI-Powered Mobile Application for Breast Cancer Prediction"** represents our original work carried out for the EPICS project as students of **VIT Bhopal University**. To the best of our knowledge, this report contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution.

The research, design, implementation, and documentation presented in this report are based on our own efforts and understanding developed during the course of this project. All external sources of information, including research papers, articles, datasets, books, websites, frameworks, and tools, have been properly acknowledged in the References section.

We take full responsibility for the authenticity of the work and confirm that it adheres to the academic integrity guidelines prescribed by VIT Bhopal University.

Date **Reg No & Name**

23BCE11456- SMRITI PRAJAPATI

23BCE11243- HRITHEMA

23BAI10178- ADITI RAJ

23BAI10341- TANMAY AGNIHOTRI

23BHI10034- GHAZIAH SHOEB

23BCE10963- AADITYA JAIN

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
OCR	Optical Character Recognition

TFLite	TensorFlow Lite
API	Application Programming Interface
UI	User Interface
UX	User Experience
SDK	Software Development Kit
IDE	Integrated Development Environment
XML	eXtensible Markup Language
JSON	JavaScript Object Notation
DB	Database
RAM	Random Access Memory
CPU	Central Processing Unit
GPU	Graphics Processing Unit
OS	Operating System
APK	Android Package Kit
NLP	Natural Language Processing
ANN	Artificial Neural Network
SVM	Support Vector Machine
Abbreviation	**Full Form**
KNN	K-Nearest Neighbors
RF	Random Forest
LR	Logistic Regression
TPR	True Positive Rate

FPR	False Positive Rate
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
CSV	Comma Separated Values
PNG	Portable Network Graphics
JPG/JPEG	Joint Photographic Experts Group
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
GUI	Graphical User Interface
DBMS	Database Management System
SQL	Structured Query Language
NoSQL	Not Only Structured Query Language
SDK	Software Development Kit
SDK API	Software Development Kit Application Interface
Bitmap	Binary Map Image Format
CI/CD	Continuous Integration / Continuous Deployment

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ABSTRACT

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. It accounts for a significant percentage of cancer-related deaths due to delayed diagnosis and lack of early detection facilities. According to global health studies, early detection of breast cancer can greatly improve survival rates and reduce mortality. However, many individuals lack access to timely medical screening and diagnostic tools, especially in resource-limited settings.

Purpose: The purpose of this project is to develop an intelligent mobile application, BreastScan, that assists users in the early detection of breast cancer using machine learning techniques. The application provides multiple prediction methods, including manual data entry, medical report analysis using Optical Character Recognition (OCR), and image-based prediction. It also includes a Quick Self-Check feature and educational resources to promote awareness and preventive care.

Methodology: The development of BreastScan follows a systematic approach using Android Studio with Java and XML for frontend development. Machine learning models are integrated using TensorFlow Lite to perform on-device predictions, while Google ML Kit OCR is used for extracting medical data from reports. Feature normalization and data preprocessing techniques are applied to improve prediction accuracy.

Findings: The BreastScan application demonstrates the ability to provide quick and efficient prediction results using multiple input methods. OCR-based report analysis reduces manual effort, and image-based prediction enhances diagnostic capability. The application improves user awareness through self-check features and educational content, making it a useful supportive tool for preliminary breast cancer assessment.

Keywords: Breast Cancer, Early Detection, Machine Learning, TensorFlow Lite, OCR, Google ML Kit, Image Prediction, Mobile Application, Health Awareness.

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CHAPTER-1

PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

Out of nowhere, a quiet threat grows - cells in the breast turn wild, multiplying without control. When left unseen, they travel, creeping into other areas of the body before anyone notices. Across the world, more women face this than many realize, year after year. Numbers don't lie: it ranks high among reasons women die from cancer. Spotting it sooner changes everything, shifting outcomes in ways that matter deeply.

Starting early makes treatment more likely to succeed, cutting down on deaths. Though tools like mammograms, tissue tests, and physical checks work well, they depend heavily on expert staff, special clinics, and high-end machines. Places far from cities or short on supplies struggle to offer these services, so delays happen - results get worse because of it.

Fast-moving tech progress opens new paths for spotting illness sooner. Because machines now learn like never before, they help find hidden signs in health records. While smartphones reach nearly everyone, apps turn into handy tools for personal care checks. Though old ways miss subtle clues, smart software catches shift across huge amounts of patient details. Even small changes in data stand out when algorithms work through symptoms over time. As devices get smarter, their role grows beyond tracking steps to watching warning signals quietly.

One goal drives the BreastScan effort - using new tech to build a phone app that helps spot breast cancer earlier. Built into it are tools like smart algorithms, text scanning from documents, plus ways to study pictures closely. Instead of just typing details, people might drop in health records so the system pulls out facts itself. Another path opens when someone adds photos instead, letting the software search for signs through visual clues.

Besides guessing risks, BreastScan focuses on teaching people what to watch for. Starting with how your body normally feels, a built-in self-check tool guides step by step. Information appears in small pieces about signs of breast cancer, why it happens, plus ways to lower risk. Knowledge grows slowly through these bits, helping users pay closer attention to their bodies. Awareness builds not all at once, but over time with repeated use. Folks might see BreastScan differently at first glance - yet it’s really about blending smart tech with everyday health needs. Rather than stepping into the doctor’s role, it steps alongside, offering help before problems grow large. Early spotting becomes easier when tools like this exist, nudging people to pay attention sooner rather than later. Better results often follow when care starts earlier, quietly shifting how things can turn out.

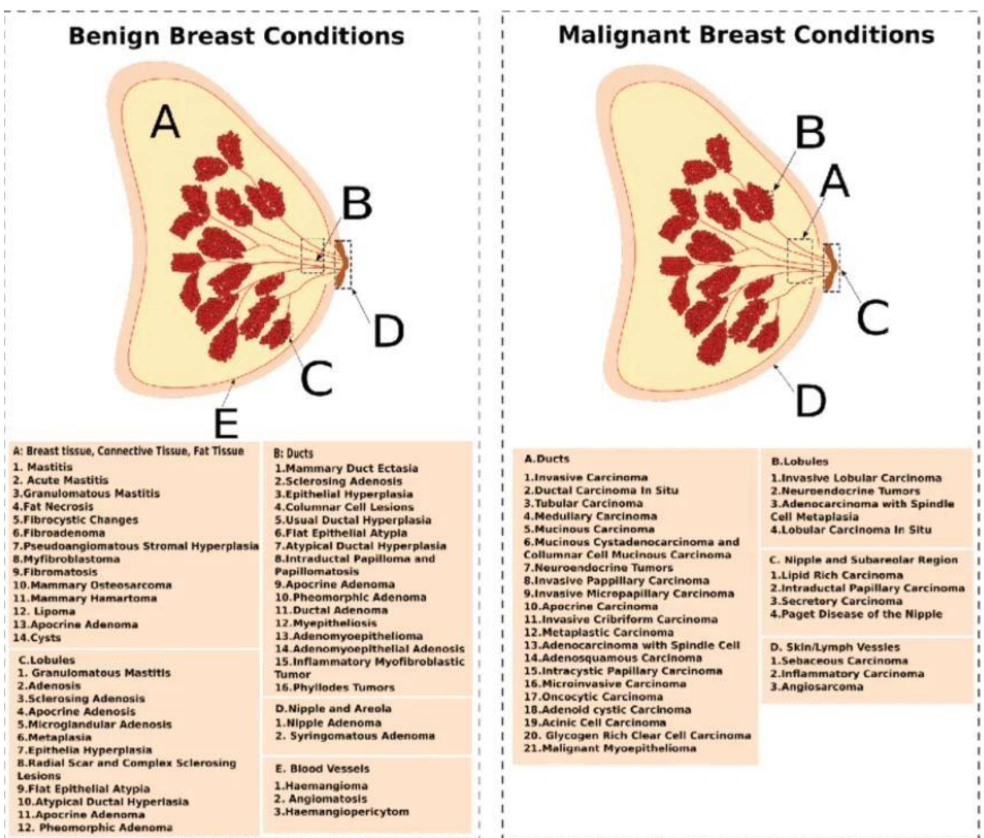


Fig. 1.1 Benign vs Malignant Comparison

1.2 Motivation for the Work

BreastScan came about because more people are facing breast cancer, yet spotting it early remains tough. Even though medicine has moved forward, too many diagnoses happen only after the disease has advanced, making recovery much harder. In places where clinics are few or far away, the problem grows even worse.

It often starts small - someone misses a sign, then another. Without knowing what to look for, changes go unnoticed. Most do not realize how crucial checking themselves can be. When warnings are ignored, time slips away. Tools could help here, if they teach while they check. Learning happens best when it fits into daily moments. Awareness grows not through lectures, but quiet reminders built into routine.

What also matters is how hard it can be to go through regular checkups, along with what they cost. Getting scans or tissue samples means trips to clinics, machines that few places have, plus help from trained staff. People living far from cities often struggle just to reach these services. With phones doing more every day, a tool built into them could offer early checks without travel or high fees.

Smartphones keep getting smarter, opening doors for health tools that fit right in your pocket. Because apps have grown so capable, they can now handle tasks once reserved for big computers. Processing power inside phones lets them run smart systems without delay. Some of these gadgets track information the moment it happens, feeding live updates to software. New ways of analyzing data make it possible to build health helpers that learn over time. These changes push developers to craft programs tuned specifically for handheld use. Efficiency matters more than ever when running complex code on small screens. With each upgrade, mobile tech becomes better at supporting personal care routines. Running intelligence locally means less waiting, fewer errors, smoother results. Today's devices do heavy lifting quietly, letting medical apps work fast and steady.

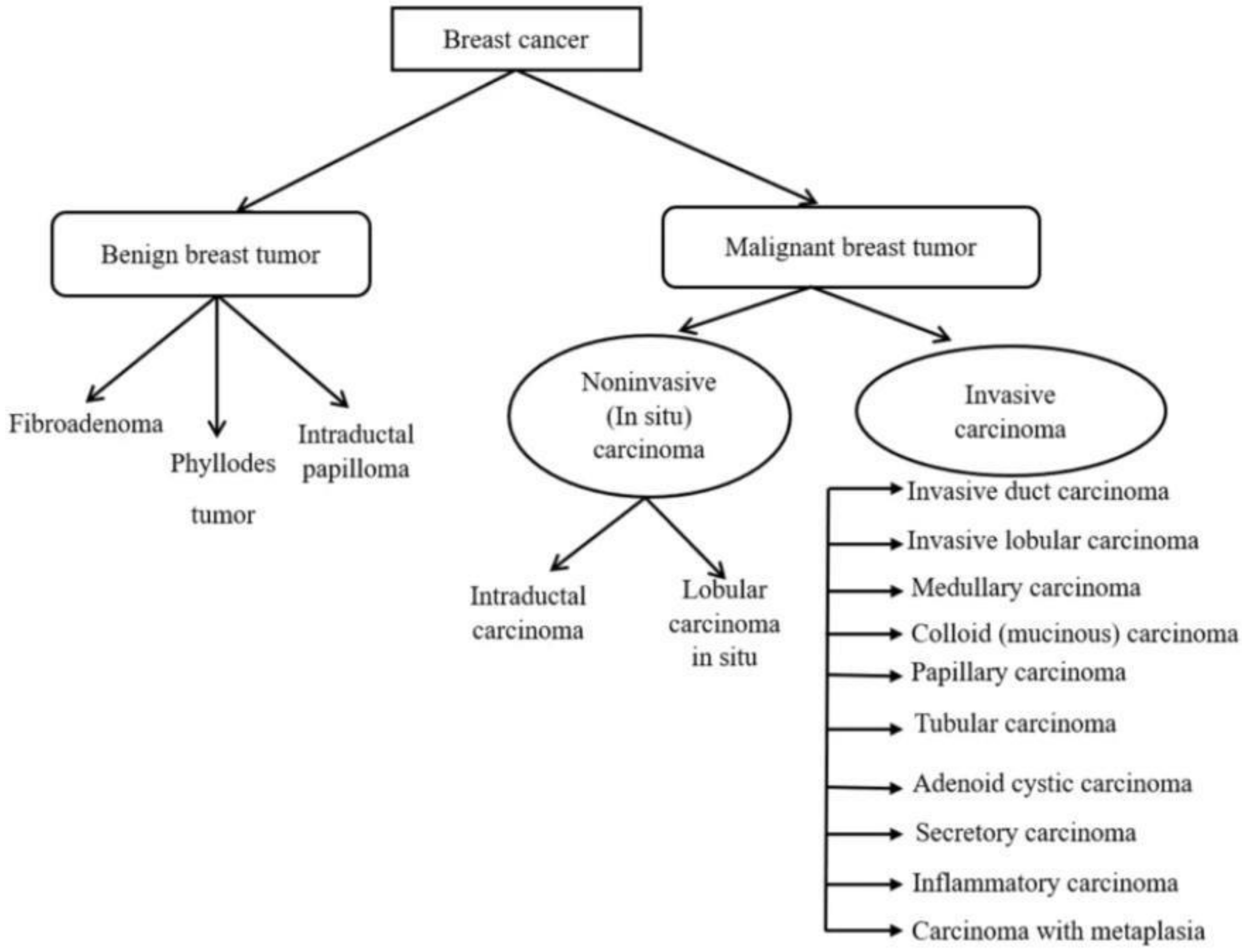


Fig 1.2 Breast Cancer Flowchart

Mobile apps now run smart prediction systems right inside them, thanks to tools like TensorFlow Lite. Running locally means no internet is needed, responses come quicker, while personal information stays put on the device. Another option, Google ML Kit, pulls useful details from paper records by reading text automatically. Medical forms get processed without manual typing, using built-in phone intelligence. These approaches keep operations smooth even when connections fail. One reason behind the design - bringing several tools together inside one app matters a lot. Instead of just picking one method, it uses predictions from forms, scans text through optical character reading, plus sorts images by pattern - all packed into BreastScan. This mix helps handle varied tasks people actually face.

Breast cancer awareness pushes this effort forward. What matters here is building something people can actually use - simple, smart tools anyone might access. Efficiency becomes key when time counts. A person gains power through clear information about their body. Early warnings shift outcomes dramatically. Control changes hands when technology serves the individual directly.

1.3 Overview of BreastScan and Techniques Used

Something that helps spot signs early - BreastScan works through your phone. Not just one feature but layers: typing in details, reading medical files automatically, even checking pictures for patterns. A tool built quietly into daily life, helping people pay attention without confusion. Instead of waiting, there's now a way to track what might matter sooner. Simple actions add up to something clear over time. No need for expertise when the app guides each step gently. Health checks become part of routine, like notes you keep or messages sent. Behind it all, calculations happen fast while keeping things private. Users stay informed because updates come straight, never exaggerated. It fits pockets, minds, moments - whenever someone wants clarity.

Starting out, the app gets built inside Android Studio. Frontend work relies on Java alongside XML for shaping how things look and function. A split structure runs through the system, each piece managing separate tasks. One part takes what users type or tap. Another shifts raw entries into usable information. Prediction happens thanks to machine learning tucked within a dedicated module. Visuals come alive after calculations finish. Each section connects without overlapping.

One key part of the setup involves a tool that reads handwritten or printed reports using smart scanning tech. Using Google's ML Kit, it pulls out visible words and numbers from any document added by the user. Once pulled, those details get scanned for specific figures needed later on. Those numbers move straight into their correct spots without typing. Less hands-on work means fewer slips during entry. Before moving forward, people can check what was taken and change anything off mark.

Picture by picture, the forecasting tool brings something extra into play. Uploading scans lets people feed data straight into the system. Once inside, those pictures get adjusted and reshaped to fit what the algorithm needs. Running on lightweight code, the analyzer sorts each one into categories. Seeing results this way opens up new ways to understand inputs. Power grows quietly behind the scenes.

Besides prediction tools, BreastScan offers a Quick Self-Check giving step-by-step guidance for simple personal checks. Users become more aware through this tool, which gently prompts routine attention to well-being. An educational part of the app shares facts on breast cancer - covering signs, origins, ways to reduce risk, along with available care paths.

Running straight on your device, it handles tasks quicker while keeping information private. Without needing outside servers, operation continues even without Wi-Fi. Access stays possible where connections are weak or missing.

From start to finish, BreastScan brings together tools like machine learning, mobile tech, yet also optical character recognition into one focused health app. Because different methods work together, the system stays sharp, adapts well, while staying simple - helping users catch warning signs earlier without confusion.

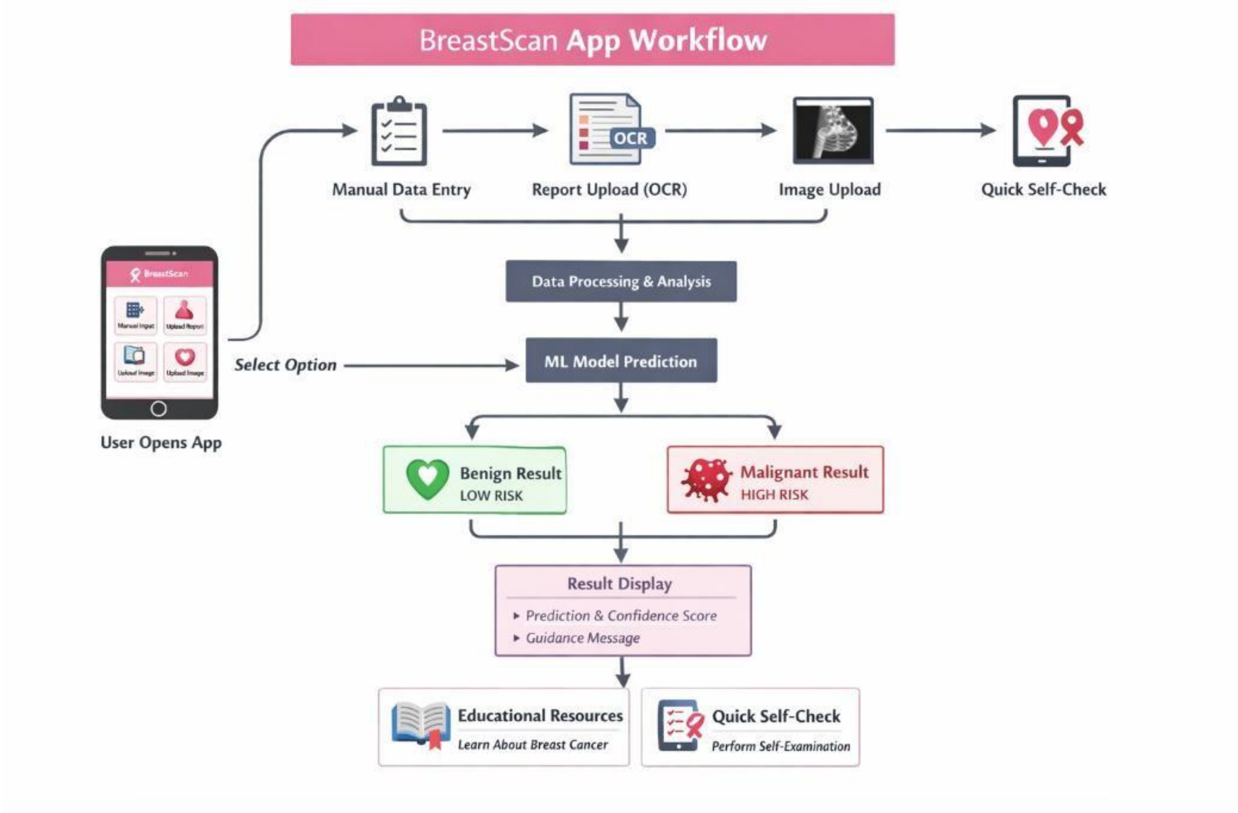


Fig 1.3 BreastScan Application Workflow Diagram

1.4 Problem Statement

Around the world, breast cancer still takes too many lives because it often shows up too late. Not everyone can reach testing centers - distance gets in the way, so does cost, sometimes roads just aren't there. High-tech tools exist, yet they sit far from those who need them most. Waiting until symptoms appear means treatment starts behind schedule. Spotting changes earlier would shift outcomes, that much is clear. When care comes slowly, chances shrink without notice.

The biggest issue is not knowing much about how breast cancer shows up or why checking early matters. Some people skip looking at their bodies often, waiting too long before seeing a doctor. When symptoms finally push someone to act, the illness might already be deeper in the body. Treatment works less well when found late, making tough outcomes more likely.

Hospital trips, expert staff, specialist tools - these old-school tests demand a lot.

When distance blocks access, getting checked becomes tough. Cost piles up quickly, slowing things down for countless patients. Delay follows delay where resources run thin. Missed checkups pile into bigger problems later on.

Mistakes happen when people read medical reports by hand. Pulling key details out takes effort, plus deep knowledge that isn't always on hand. The diagnosis gets harder because of it - sometimes results go off track.

The only reason for these tools to fall short is that they rarely bring together multiple ways of diagnosing onto one stage. While some handle number patterns well, others deal only with visuals - rarely do they mix the two. Because of this split, performance often stumbles, so does how easily people can use them.

Worries around keeping health information private pop up a lot these days. Sending patient details online happens often because many tools depend on remote servers. When data moves across networks, someone might grab it without permission. That chance makes people uneasy about safety.

A working solution must be clear, safe, easy to reach, plus help catch breast cancer sooner. This tool could combine different ways to examine data, rely less on outside tools, while also helping people understand risks.

Early signs matter most. This tool moves where people are, not stuck in clinics. Learning from data shapes how it sees patterns. Words on paper turn into digital insight through smart scanning tricks. Seeing images clearly helps spot changes fast. Help comes before crisis strikes. Simplicity guides every screen shown. Real needs shape its design, not trends. Quiet progress beats loud promises here.

1.5 Objectives of the Work

BreastScan wants to build a smart phone app that helps catch breast cancer sooner through tools like machine learning instead of just relying on standard methods. Built around ease of use, it pulls together image scanning tech plus decision-making algorithms behind the scenes. Accuracy matters here - results should reflect real patterns without guesswork entering the picture. Speed plays a role too, cutting down wait times compared to traditional pathways. Access counts more than ever, bringing screening support straight into everyday hands regardless of location. Help comes quietly, embedded within familiar devices rather than clinical settings. No flashy promises - just steady analysis when someone needs insight fast. Design leans toward clarity so confusion fades during moments that matter most. Predictions form based on data shapes only, avoiding assumptions along the way. Each part works because others do their job right, unseen but necessary.

Instant results come from entering clinical details into forms, a main goal here. Input by hand means health insights show up fast - no need for hospital-grade tools nearby. The key aim is to turn medical report reviews into something automatic through OCR tools. When reports get uploaded, useful details are pulled out without needing someone to type them. Because less handwork happens, things move faster around here. Mistakes that sneak in when people enter data by hand tend to drop off too. Pictures help too - users can send in scans so the tool gives better guesses about health issues. Because of that, checking symptoms gets sharper and more reliable. Awareness takes center stage here too. Inside the app, a fast self-check tool appears alongside clear guides about breast cancer - what signs look like, where it comes from, because knowing matters. Spotting changes sooner becomes more likely when information sits close at hand. Early notice grows out of small habits done regularly. Starting simple keeps things clear. A smooth way to move around the app comes first, so finding tools feels natural instead of confusing. Phones work just as well as bigger screens, opening the door for more people anytime, anywhere.

A key goal means keeping user information private through local computation. Instead of sending details elsewhere, analysis happens right on the device. With TensorFlow Lite, functions work even when disconnected from the internet. Sensitive inputs never leave the hardware they start on.

One last thing - this effort shows what happens when smart machines step into medicine. Not just combining tools, but linking them tightly inside one platform called BreastScan. This emerges glimpse at fresh ways tech might help more people stay healthy. Instead of waiting around, the system puts ideas into motion where they matter most.

1.6 Organization of the Project

This document unfolds across several sections, every one zooming in on a different piece of the BreastScan initiative. Built step by step, it walks through how the idea took shape, moved forward, then got tested along the way.

A look at Chapter 1 opens the door to what follows, covering context alongside reasons behind the work through clear aims woven into a defined layout. This part plants early seeds so readers grasp why it matters, how far it reaches.

Looking at what others have done comes first in Chapter 2. Past studies on spotting breast cancer get examined here, yet some methods fall short in practice. Old tech shows gaps that still remain unresolved. Because of these flaws, a new approach feels less like an option and more like a necessity. The reasons behind building something different start making sense halfway through.

What you find in Chapter 3 revolves around the tools and rules a system needs to work - things like what kind of computer power it demands, which programs must run, what tasks it should handle, how fast it ought to respond. Each detail drawn out shapes how well everything will come together later on.

How the pieces fit together shows up early in Chapter 4. Instead of just listing parts, it walks through choices behind the layout. One piece connects to another, then another, building slowly. Each section does its job while linking outward. The way things are arranged reveals thinking over time. Structure here isn't random - patterns guide each step. Seeing how modules talk helps make sense of the whole.

A look at hands-on setup begins here, diving into code structure alongside model training steps. Following that, reading scanned text gets handled through automated recognition tools. Once pieces connect, trial runs check whether everything works as expected. Theory takes shape when put into real use during these tests.

Outcomes show up in Chapter 6, along with where they might actually be used. Key findings take center stage here, followed by how well the system ran. Real-life uses pop up next, tied closely to what was achieved. Performance details linger throughout, grounding claims in observable behavior.

Wrapping things up, Chapter 7 pulls together what's been done so far while peeking ahead at possible upgrades. Some shortcomings of today's setup come into view here, opening space for new ideas later on.

1.7 Summary

A look at the BreastScan initiative opened this section, showing why it matters along with its goals and how it is set up. Right away, attention turned to spotting breast cancer sooner, underlining how tech helps tackle tough medical issues.

A fresh look at why things were built this way kicked off the conversation - clarity and speed mattered most. What came next shifted into how it actually works, sketching out tools and parts that make the whole thing run.

This chapter lays the groundwork for BreastScan, opening space for what comes next. What follows digs into how it takes shape, gets built, then changes things. Each part moves forward from here.

CHAPTER-2

EXISTING WORK

2.1 Introduction

Spotting breast cancer early matters because it affects so many people worldwide. Because lives depend on timely diagnosis, scientists keep searching for better tools. Instead of relying only on human eyes, computers now help spot warning signs faster. Thanks to smart algorithms that learn patterns, machines can highlight suspicious areas in scans. Over time these systems get sharper at telling normal tissue apart from possible tumors. Doctors gain support through tech that flags concerns without replacing their judgment.

Looking at old ways to find breast cancer, doctors usually check patients physically, use X-ray scans, sound wave pictures, or take tissue samples. Even though those tools work well, they need expert staff, special machines, plus plenty of time to review results. In villages or areas with few resources, getting hold of such care can be tough - sometimes impossible. Late diagnoses pop up more often when access is limited. Because of this gap, scientists now build smart computer helpers meant to catch signs earlier.

Lately, research teams-built tools using machine learning to guess breast cancer risks from patient details. Not far behind, deep learning methods like convolutional networks took off analyzing scans instead of numbers. Reading doctor sometimes notes involves systems that extract information directly from paperwork by converting images into text. Each method operates differently, with its own strengths and limitations, making careful evaluation essential.

Looking closer at how breast cancer detection works today shows many methods already in play. Some rely on imaging; others blend data from different sources. One after another, these strategies show strengths - yet none quite bridge every challenge.

What stands out is a pattern of partial fixes instead of full answers. Each method misses something small, sometimes critical. Gaps appear when speed meets accuracy under real conditions. That space between expectation and result shapes new thinking. Enter BreastScan, built not to replace but to connect. Instead of chasing one breakthrough, it links what exists into a smoother path. Past efforts help map where progress stalled. From those pauses come clearer goals. This section walks through those steps without rushing ahead. Progress hides in details often overlooked. Seeing them changes where you look next.

2.2 AI used to find breast cancer

Now showing up in clinics, artificial intelligence helps spot health issues like breast cancer faster than before. Instead of just reading charts by eye, computers can now sort through piles of information, noticing tiny clues people might miss. When it comes to finding tumors early, these tools examine scans, doctor notes, and past cases all at once. What stands out is how precisely they highlight warning signs others could overlook. Behind the scenes, machine learning digs into details across thousands of examples, building smarter guesses each time. Because every case differs slightly, the system adapts without needing explicit reprogramming. Over months, performance grows more reliable, catching irregularities sooner. While humans still make final calls, support from algorithms sharpens their view. This blend of tech and expertise opens paths to earlier interventions. Yet results depend heavily on data quality, not just speed. Surprising shifts happen when models learn from diverse populations, reducing blind spots. Even so, errors linger if training samples lack variety. From start to finish, progress hinges on balance - between automation and judgment.

One way AI helps spot breast cancer is through machine learning. From examples, programs learn to sort findings - calling them harmless or dangerous. Support Vector Machines do this job, also Decision Trees, plus others like Random Forest and

Logistic Regression. Instead of guessing, they follow clues found in medical records: how big a lump is, its outline, surface feel. Patterns in numbers guide their conclusions, shaped by earlier cases marked right or wrong.

A prominent development in AI for healthcare is the integration of natural language processing with optical character recognition, enabling systems to extract and interpret detailed information from written medical records. These systems turn messy notes into clean layouts ready for review. Because machines handle the pulling now, people spend less time typing things over - mistakes drop too. Still, how well artificial intelligence works ties closely to the kind and amount of information it learns from. At the same time, worries around keeping personal details private and systems secure need clear answers before they can run without risk. From the start, artificial intelligence helps spot breast cancer more effectively today while shaping smart tools such as BreastScan along the way.

2.3 Existing Approaches/Methods

Over time, different ways to spot and forecast breast cancer with computers have come about. While some rely on machine learning, others lean heavily on deep learning applied to images. A few take a different path altogether - pulling data from medical reports using OCR tools. Each method stands apart, yet targets the same goal through separate lanes.

Looking at organized medical records is what machine learning techniques usually do. Training steps happen using information like how tumors appear, past patient details, together with test results. After learning patterns, systems sort fresh cases as noncancerous or cancerous. Simplicity marks these methods, also they run efficiently without heavy computing needs - ideal when speed matters during active use.

Pictures of paperwork get turned into digital words by machines that read them like people do. Because of this, hospitals spend less time typing things out by hand. The method swaps old-school clerical work for quicker electronic handling. Machines see the letters on scanned pages then write them down again in a format computers understand. This helps clinics move faster without slowing down.

Though good in their own ways, every method comes with drawbacks. When faced with messy inputs, machine learning can fall short. Deep learning needs massive amounts of data plus serious computing power. Fuzzy pictures tend to trip up OCR systems.

A fresh take on detection begins with combining methods under one roof - BreastScan builds a full picture without leaving gaps. Ways once separate now link, forming a steady path toward spotting risks early.

2.3.1 Machine Learning-Based Prediction

One way people spot breast cancer uses machine learning. Not every method works the same, but this one learns patterns from old records to guess outcomes for fresh patients. Training happens on information already sorted into noncancerous and cancerous growths. What comes out depends heavily on how clear those earlier labels were.

One popular method relies on tools like Support Vector Machines, then there are Decision Trees that split data step by step. Tumor size guides classification, yet shape adds extra clues while texture gives more depth alongside clinical details. What stands out is how certain inputs get picked carefully before processing begins. Accuracy climbs when noise fades early in the pipeline, just as wrong choices can drag performance down later.

What makes machine learning predictions stand out is how straightforward they are. Despite their low demand on computing resources, they still run smoothly inside phone apps - thanks to tools such as TensorFlow Lite.

Breast cancer forecasting still leans heavily on machine learning, even with hurdles standing in the way. Built into countless smart medical platforms, it plays a central role despite ongoing difficulties.

2.3.2 Image Based Deep Learning Models

Suddenly, deep learning built on pictures changed how doctors' study medical scans. Instead of traditional methods, these systems rely on special networks that scan visuals to spot signs of illness, like tumors in breast tissue. What makes them stand out is their knack for catching shapes and surface details across images. Because of this, they work well with X-ray views of breasts or microscope samples of body tissues.

From raw pictures, deep learning pulls out key details without human help. Because of this, it often scores better than older techniques. Tumors show up more clearly when CNNs scan mammograms. Sorting image types is one task they handle well. Even marking exact sick areas becomes possible through their analysis.

Picture-focused deep learning handles tough information while delivering sharp insights. Because it spots tiny details in visuals – one's people might miss - the technology boosts how correct diagnoses can be.

Yet training deep learning models demands vast amounts of data along with strong computing power. It often takes a long time to train them properly. Putting these systems onto phones brings its own set of difficulties. Image quality plays a big role in how well they work. Blurry or grainy pictures might lead to mistakes in results. Even with their flaws, hospitals today lean heavily on image-focused deep learning tools when diagnosing complex conditions. These models often shape critical medical decisions behind the scenes.

2.3.3 Reading medical reports with OCR

From medical reports, OCR tools pull out words and turn them into organized formats. Because they handle routine typing tasks, less time gets spent on handentering details.

When spotting breast cancer, doctors' notes usually hold key facts like lab outcomes, size readings, one after another written down. Machines that read text grab those pieces so algorithms later sift through them.

Picture a phone reading paper like a person - Google ML Kit makes that happen. When you snap a shot of a form, it digs out words fast. Accuracy sticks close, even if the lighting wobbles. Information jumps from page to app without tripping. It works behind the scenes, quiet and sharp.

The key advantage of OCR-based analysis is its ability to convert unorganized, unstructured information into clean and usable data. Because of that, work moves faster while mistakes by people happen less often. Another thing - it opens the door to dropping reports straight into systems without typing everything out.

Still, poor image quality can trip up OCR tools - especially if handwriting is involved. When the scan looks messy, mistakes start creeping into the extracted words. Layouts that twist and turn confuse the process even more. These hiccups ripple through the whole system, nudging error rates higher.

Even with its hurdles, reading text through OCR plays a key role in today's medical tools. BreastScan works better because it can pull data from scanned documents.

2.4 Pros and Cons of Existing Methods

Approach	Pros	Cons

Machine Learning-Based Prediction	- Provides fast and efficient prediction using structured clinical data - Requires less computational power compared to deep learning - Can be deployed easily on mobile devices using TensorFlow Lite - Works well with numerical datasets and predefined features	- Depends heavily on quality and completeness of input data - Requires manual feature extraction. - May not capture complex patterns in data. - Accuracy may be lower compared to deep learning models

Approach	Pros	Cons
Image-Based Deep Learning Models	- High accuracy in detecting patterns from medical images - Automatically extracts features from raw images - Useful for detecting tumor and abnormalities visually - Reduces human effort in image interpretation	- Requires large datasets for training - High computational and memory requirements - Performance depends on image quality - Difficult to deploy on low-end mobile devices

OCR-Based Medical Report Analysis	- Automates extraction of data from medical reports - Reduces manual data entry effort - Improves efficiency and speed of processing - Useful for handling unstructured data	- Accuracy depends on image clarity and format - Struggles with handwritten or low-quality text - May extract incorrect or incomplete data - Requires user verification for reliability
Traditional Diagnostic Methods (Mammography, Biopsy)	- Highly accurate and clinically reliable - Performed under professional medical supervision - Widely accepted in healthcare systems - Provides detailed diagnostic information	- Expensive and time-consuming - Requires hospital visits and specialized equipment - Not easily accessible in rural areas - Delays in diagnosis may occur
Mobile-Based AI	- Easy accessibility through	- Not a substitute for professional
Approach	**Pros**	**Cons**

Applications (like BreastScan)	smartphones - Provides quick and real-time predictions - Supports multiple input methods (form, image, OCR) - Promotes awareness and early detection - Works offline using on-device ML	diagnosis - Accuracy depends on model and input quality - Limited by mobile device performance - Requires continuous updates and improvements

Table 2.1 Pros and Cons of Existing Methods

2.5 Issues and Observations

Looking at current methods shows a few clear problems. What stands out most is how hard it is to get high-level testing tools. Some setups need special gear or support systems, things that aren't within reach everywhere. A big gap remains where resources are limited.

One thing stands out: good results need clean information. When machines learn, they rely heavily on correct labels and precise details instead of guesses. If what they get is messy or wrong, their answers go off track just as fast. What goes in shapes what comes out - flawed inputs lead to flawed thinking.

Most current setups skip clear layouts altogether. Without straightforward design, average people often find them tough to handle - keeping usage low.

Facing these problems shows how crucial it is to have a system that connects smoothly, stays within reach, yet keeps data safe - something the BreastScan initiative steps in to handle.

2.6 Summary

A close look at current methods for spotting breast cancer filled this section. With AI leading some efforts, machines began recognizing patterns through learning techniques. Some systems even used deep networks capable of processing images like a radiologist might. Text captured from medical records got converted into data using optical character recognition tools. Each technology brought something different to how doctors approach diagnosis.

What stood out was how each method brought its own mix of pros and cons. Challenges like spotty data, hard access, or trouble linking systems showed up clearly. Yet, what came from this gave solid ground to build BreastScan on. A fresh look begins here - what it takes to build the suggested system comes into view. Details unfold slowly, shaped by need rather than guesswork. Following through means paying attention to structure, not just goals. Every piece fits only when rules are clear. What shows up depends on careful planning ahead of time.

CHAPTER-3

REQUIREMENT ARTIFACTS

3.1 Introduction

Figuring out what people actually need kicks off nearly every solid software build. Not until teams sit down to listen do real patterns start showing up. Each detail captured shapes how code eventually behaves behind the scenes. Miss a key point early, things unravel fast later on. Well-put-together notes keep builders aligned without constant rework. Expectations set at the front often decide satisfaction levels at delivery. The whole effort leans heavily on who gets heard during planning rounds. Clarity upfront means fewer surprises when it finally runs live. Getting it right shifts energy toward building instead of fixing. Without this step, even clever coding feels misplaced.

When building the BreastScan tool, spotting what the system must do comes first - technical details, features, tasks, along with hidden needs shape early decisions. Because this app uses machine learning and optical character recognition to help predict breast cancer, knowing exactly which tools, skills, stay crucial during setup. Here comes what the BreastScan setup needs. Hardware bits sit alongside software ones, each playing its part. Data must meet clear standards so results stay on track. Functions unfold step by step, built for real tasks doctors face daily. Speed matters, yet stability matters more when decisions hang in the balance. The way people interact with screens shapes how fast they adapt. Every detail set down keeps things running without hiccups. Accuracy gets locked in through careful design choices early on. Ease of use weaves into every piece, not tacked on at the end. Smoothness from start to finish makes the difference nobody notices - until it's missing. Running on phones means screen sizes differ, so fitting everything matters. Because tasks need quick responses, how fast the processor works plays a role too. Memory becomes tight when handling several jobs at once, especially with heavy tools inside. Since reading text from images is part of what it does, software must handle image analysis well. Tools chosen have to keep up with smart algorithms that learn over time, not just static rules. Each piece fits together only if the foundation supports constant changes underneath.

The one thing that often gets overlooked is that how dependable the system must be, along with its safety. Since personal medical details are involved, guarding information becomes unavoidable. Protection steps aren’t optional here - they’re built into how things work.

This chapter sets the stage for building the BreastScan system through clear requirements and defined limits. With those in place, every piece fits both goals and what users actually need.

3.2 Hardware and software requirements

Hardware plus software must support BreastScan during building and when live. Building needs differ from what users' need on their devices.

Component	Requirement	Details
Mobile Device	Smartphone (Android)	Minimum Android version 6.0 or higher to support application features and libraries
Processor	Multi-core Processor (Snapdragon / MediaTek / equivalent)	Ensures smooth execution of machine learning models and OCR processing
RAM	4 GB or higher	Required for running TensorFlow Lite model, OCR processing, and multitasking
Storage	Minimum 16 GB free storage	Used for storing application data, model files, and temporary processing data
Component	**Requirement**	**Details**
Display Unit	Smartphone Screen	Touch-enabled interface for easy navigation and interaction with the application
Camera	Integrated smartphone camera	Required for capturing medical reports and images for OCR and prediction

Battery	3000 mAh or higher	Supports extended usage during image processing and prediction tasks
Operating System	Android OS	Required platform for application deployment and execution
OCR Tool	Google ML Kit	Used for extracting text from medical reports and images
Build Tool	Gradle	Manages dependencies and builds the Android application efficiently
Testing Environment	Android Emulator / Real Device	Used for testing application performance, compatibility, and usability

Table 3.1 Hardware and Software Requirements

3.3 Specific Project Requirements

What the BreastScan system must do comes down to clear details about how it manages data, what it can perform, how fast it runs, along with how people interact with it. Built this way, the goals stay on track while users move through tasks without stumbling blocks in their path.

A single way of entering info won't work here - typing it in works, but so does sending reports or pictures. One after another, these inputs get handled quickly, without delays that slow things down. Right answers come out because the steps inside make sense. People using it learn as they go, guided by what shows up when needed.

Breaking down the project needs comes next - data demands show up first, followed by what the system must do. How well it runs matters just as much, tied closely to safety checks built in along the way. The look and feel users interact with stands separate but linked. Every piece fit into how the whole thing operates in the end.

3.3.1 Data Requirements

Picture this: without solid info, BreastScan can't do its job well. What it sees matters - numbers on a chart, scans on screen. One feeds facts, the other shapes meaning. When details are messy, results waver. With clear inputs, clearer answers follow. It runs on two kinds of records - one measured, one visual. Either piece missing throws off balance. Precision hides in how things are gathered. Not just what's there - but how it sits together.

Numbers pulled from medical scans - like how round a spot is, its surface feel, outer edge size, total space it covers, and evenness - are typical details found in records of breast health cases. Because machines learn from these figures, they need to arrive clean and scaled right. Without that setup step, guesses made later could go off track. A hiccup in formatting might tilt results sideways.

Picture this: medical papers filled with numbers and findings go into the system. Not just any images, but ones showing lab outcomes or scanned notes. From there, a tool reads the words off each page. Instead of leaving them buried in pictures, it pulls out key details. Then comes transformation - plain text becomes organized entries. What once was locked in scans now lives as usable records.

A single picture can set off a chain inside the system, where scans get reshaped into uniform sizes ahead of analysis. Sharpness, contrast, even lighting - each detail quietly shapes how well the model guesses what lies within.

After predictions wrap up, temporary data gets wiped clean. Information that could identify users never sticks around inside the setup, keeping things safe plus private. Processing runs on short-lived inputs - gone once duties finish.

3.3.2 Functions Requirements

Functionality	Description
User Authentication	Allows users to register, log in, and securely access their profile and application features
Form-Based Prediction	Enables users to manually enter clinical parameters and get prediction results
OCR Report Upload	Allows users to upload medical reports and automatically extract data using OCR
Image-Based Prediction	Allows users to upload medical images for analysis and prediction
Data Preprocessing	Normalizes and processes input data before feeding it into the machine learning model
Prediction Output	Displays results as benign or malignant along with confidence score
Quick Self-Check	Provides guided steps for self-assessment and awareness of symptoms
Educational Content	Provides information about breast cancer, symptoms, causes, and prevention
User Profile Management	Allows users to view and update personal details
Navigation Menu	Provides access to profile, info, privacy policy, terms, and logout options

Functionality	Description
Error Handling & Validation	Ensures proper input validation and displays appropriate error messages

Table 3.2 Functions Requirements

Explanation:

Logging in lets people enter the app safely, using their own account details. Getting started means setting up a profile that remembers individual preferences. Access opens only after confirming identity through secure steps.

Starting with a form, people type in health details by hand - then see outcomes right away. A quick process swaps numbers for estimates without delay. Type it once, get answers fast. Filled fields lead straight to forecasts. Input turns into insight nearly at once.

Upload a report so the text gets pulled out by scanning it right away. Machines grab key details without typing. Picture becomes data through smart reading tools. Info moves fast when documents are snapped into digital form.

Pictures get checked by smart software when someone sends them in. One way it works is through learning systems that study visuals. These tools guess what an image shows after looking at its details. A person can share a photo so the program examines it closely. The system uses past examples to understand new pictures uploaded.

Built into the system, data preprocessing cleans up information before analysis. It adjusts values so patterns stand out more clearly. Smoothing differences happens early, making results line up better later. Shaping inputs this way supports stronger guesses down the road. Consistency shows when outputs reflect cleaner starting points.

Results show up clean. Classification appears alongside how sure the system is. Clear layout makes it easy to grasp what came out.
A moment to pause. Spot subtle hints of trouble before they grow. Step-by-step guidance shows what to watch for. Clues add up quietly - this makes them clearer.
Notice small shifts on your own time. Early signals gain shape when broken down. Guidance simplifies what feels vague.
Breast cancer knowledge grows when people learn what signs to watch. Awareness spreads through clear information on how to stay safe. Learning steps that help prevent illness becomes easier with straightforward teaching. Knowing more leads some to take earlier action. Simple facts shared widely make a difference over time.
A person can update their info using the app. Their name, email, or picture stays under their control. Changes happen right inside the feature. What they put in is what shows up elsewhere. Editing takes just a few taps. Info moves across screens when saved. Each detail remains theirs alone.
10.Navigation Menu: Provides easy access to different features and settings. Built-in checks catch mistakes before they cause trouble. Invalid entries get blocked automatically, keeping things running smoothly. Mistakes at the start never slip through. System stays stable because bad data stops early. Wrong inputs do not pass go.

3.3.3 Performance and Security Requirements

Performance	Security
Real-time prediction generation within 2–3 seconds	Secure user authentication and login validation
Fast OCR text extraction from reports	No unnecessary storage of sensitive medical data

Performance	Security
Efficient image processing with minimal delay	On-device processing using TensorFlow Lite
≥ 90% accuracy in prediction (depends on model)	Protection against unauthorized access
Smooth app performance on mid-range devices	Data privacy maintained without cloud dependency
Minimal latency in UI response and navigation	Secure handling of input and temporary data

Table 3.4 Performance and Security Requirements

Performance:

Right now, it shows predictions almost instantly - just a couple of seconds pass. A brief wait, then answers appear like clockwork. Speed stays steady, never dragging behind.

Text gets pulled fast from uploaded files using OCR. That happens right after upload, turning images into editable words almost instantly.

Pictures help guess what happens next on phones. Devices use them faster now when scrolling through apps.

Folks might see results near 90% when the data going in is clean. Still, it shifts a bit based on how well that info holds up. Rough inputs often lead to lower scores. Performance dips show up if details are missing or messy. Good signals help keep predictions closer to true outcomes. Each run changes slightly even with similar starting points.

Faster response times come from smart coding built into everyday phones. Smooth motion stays steady even when screens fill up. No hiccups happen during normal use because systems handle tasks quietly. Efficiency grows behind the scenes while users tap through menus.

Security:

Signing in safely begins when the system checks who you are. That check decides what parts of the app open up next. Only your profile unlocks specific areas others can't reach.

By design, it doesn't keep private health records around long term.

Predictions happen right where you are - inside the device - so nothing leaves your hands. Privacy stays intact because everything runs locally, without a detour. Your data does not travel; it simply works where it belongs.

Fresh code guards against intrusions by design. Built-in checks block outsiders before they get close. Safety lives deep in how it runs. Every layer resists tampering quietly.

Once tasks finish, any short-term information created along the way gets removed right away. This keeps personal details out of long-term storage by design. Files that exist just for calculations vanish as soon as they're no longer needed. The system does not hold on to fragments once operations complete. Sensitive bits disappear automatically when steps wrap up.

3.3.4 User Interface Design Standards

How users interact with BreastScan affects how well they can use it. A clean layout helps people understand things fast, because clarity matters when navigating screens. Looks count too - when colors and spacing feel balanced, attention stays focused without effort pulling away. Smooth flow from one step to the next keeps frustration low during tasks. Enjoying the look often means staying longer, even if that was never planned at first.

A fresh look on the screen helps people move around without trouble. Moving through sections feels smooth when labels make sense. Colors that fit together guide eyes where they need to go. Text shows clearly when sized right and spaced out properly. Small pictures in the interface point users along without words. Each piece sits where it belongs so nothing seems messy. Seeing everything at a glance cuts down mistakes. Style supports function instead of getting in the way.

Easy-to-use input fields need straightforward labels plus helpful error notes. Right spot for buttons makes moving through the interface feel natural. Outcomes show up clean, sharing predictions along with how certain they are.

A shift in screen size that changes how the interface behaves. When orientation flips, the layout adjusts without skipping a beat. Different devices handle it differently, yet the experience stays steady. Performance holds firm no matter the gadget in hand.

3.4 Summary

A close look at what BreastScan needs was laid out here. Hardware specs came into view, followed by the software side. Data demands were outlined alongside tasks the system must handle. Performance traits took shape, each detail fitting into place. Design choices for how users interact showed up last.

A solid base starts here, shaping how the system comes together through design and build. User goals come first - efficiency matters just as much, while keeping information safe stays nonnegotiable.

A solid grasp of these needs shapes how the work moves forward - step by step, piece by piece. Following this, attention shifts to how BreastScan takes form, its blueprint and inner structure unpacked in what comes next.

CHAPTER- 4

DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

Building the BreastScan app uses a clear step by step plan split into separate parts, helping keep things running smoothly while allowing room to grow, stay precise, and make updates without hassle. Instead of one long build, work moves forward in loops - each round adds new pieces, checks performance, takes user thoughts into account, then improves what was done before. Because medical tools demand high precision, going through cycles helps catch issues early, making sure results remain trustworthy throughout.

Right off the bat, teams look at what users actually need. Instead of guessing, they dig into the problem area to see how things work now. From there, features start taking shape based on real tasks the system must handle. Performance rules show up too - speed, reliability, limits - all spelled out early. One feeds into the next, so nothing gets missed down the line. Without this step, later stages wobble from the start. Clarity here means fewer surprises when building begins. It sets the pace without calling attention to itself.

Coding begins once plans are set, shaping each piece of the app step by step. User screens take form through XML, built one section at a time. Java handles how things work behind the scenes, directing actions as needed. Smart features come alive with TensorFlow Lite weaving in learning tools quietly. Text grabs from medical papers happen via Google ML Kit, slipping words out of images. Parts grow apart, then meet up when ready, fitting together like puzzle pieces late in the process. Pieces click into place only after testing shows they stand on their own first.

Right after design comes checking how well things actually work. That stage puts the setup through real checks on speed, function, correct output. Different methods

- like examining single parts, linking modules, full run-throughs - help spot hiccups early. When issues show up, notes from testers guide updates right away. Because changes feedback fast, each version gets noticeably better without extra steps piling on.

One thing stands clear - spotting breast cancer early matters. This app works by studying patterns through smart software. Not everyone knows medical terms, so it keeps things simple. What helps most is how anyone can use it, no special training needed. Predictions come from data, shaped by learning algorithms behind the scenes. A phone becomes a tool, putting insight into everyday hands. Easy steps guide each move, avoiding confusion. Clarity stays central, even when handling complex health signals. People gain awareness without needing to be experts. Technology bends toward real needs here, quietly supporting care.

From start to finish, the approach holds up well under pressure while staying lean. It grows smoothly when demand rises, thanks to built-in flexibility. Each part moves toward the goal - making health care easier to reach and spotting issues sooner - not through force, but fit.

4.2 Functional Modules Design and Analysis

Module	Objective	Tools	Operation
User Authentication Module	Capture and authenticate user details securely	Firebase Authentication / Local Storage	Enables user registration, login, and profile management
Form-Based Prediction Module	Allow users to input clinical data for prediction	Android SDK, TensorFlow Lite	Takes user input, preprocesses data, and generates prediction results

Module	Objective	Tools	Operation
OCR Report Analysis Module	Extract medical data from reports automatically	Google ML Kit OCR	Scans uploaded reports, extracts text, and autofills input fields
Image-Based Prediction Module	Perform prediction using uploaded images	TensorFlow Lite, Android Bitmap Processing	Processes images and predicts whether tumor is benign or malignant
Data Preprocessing Module	Prepare and normalize input data	Java (Custom Logic)	Applies normalization and formatting before passing data to ML model
Prediction Engine Module	Generate prediction results	TensorFlow Lite Interpreter	Runs ML model and outputs prediction with confidence score
Quick Self-Check Module	Provide awareness through guided self-check steps	Android UI Components	Displays instructions and questions for selfassessment
Educational Content Module	Provide information about breast cancer	Static Content (Text/Images)	Displays symptoms, causes, and prevention details
User Profile & Menu Module	Manage user navigation and profile data	Android SDK	Provides access to profile, info, privacy policy, and logout

Result Display Module	Show prediction results clearly	Android UI Components	Displays benign/malignant result with confidence score
Module	**Objective**	**Tools**	**Operation**
Error Handling Module	Handle invalid inputs and system errors	Java Exception Handling	Validates input and shows appropriate error messages

Table 4.1 Functional Module Breakdown

4.2.1 Form-Based Prediction Module

Inside BreastScan, a key piece sits quietly - the form-based prediction tool. Inputting details by hand happens here, focused on signs tied to breast cancer. Starting off, the layout offers clear spots to type, along with written cues and alerts when something's wrong. Because accuracy matters, it watches every entry closely. Missing details get flagged right away. Wrong styles in dates or numbers are caught instantly. Even if someone tries a value too high or low, it won't slip through. Alerts pop up only when needed, keeping things smooth. Clarity stays front and center throughout the process. After processing, the system shows the outcome to the person, together with a measure of how certain it is about that result. Presented plainly, the information lands without confusion, so understanding comes naturally.

Most people can use this tool without trouble if they work with medical records. Quick answers come even when there is no online connection around. Light on system resources, it runs smoothly where bigger tools might struggle. Built for speed, yet gentle on hardware demands during operation.

4.3 Software Architectural Designs

Starting at the core, BreastScan uses a step-by-step structure where each part handles one job. Because functions are split apart, changes in one area don't mess up another.

Working this way keeps things tidy, easier to fix, and ready to grow when needed. Running across the setup are three key levels –

- Presentation Layer
  This layer comprises the user interface elements such as screens, buttons, and forms. It is designed with a strong focus on usability, ensuring that navigation is intuitive and user interactions are smooth. Careful attention to layout and design helps guide user actions effectively.
- Processing Layer
  The processing layer handles the core logic of the application. It manages data flow between the presentation and data layers, processes user inputs, and ensures that operations are executed efficiently and accurately.
- Data Layer
  This layer is responsible for data storage, retrieval, and management. It ensures that information is handled securely and reliably, maintaining both temporary and persistent data as required by the application.

Overall, this layered architecture promotes clarity, modularity, and scalability, enabling efficient development, testing, and future enhancements.

4.4 Summary

This part walked through how BreastScan was built, breaking down its structure and design path. From early sketches to working parts, one step followed another with clear purpose. Built in loops of testing and refining, the process stayed focused on getting results right. Each version improved upon what came before, shaping both function and form. Through repeated cycles, precision became easier to reach. The overall setup took shape gradually, guided by real feedback and practical needs. Breaking things into pieces came up a lot during talks about how to structure the system. One-piece deals with just one job, built so everything runs without hiccups.

Focused tasks inside separate blocks make the whole thing work better. Thought went into keeping parts apart but working together when needed.

This chapter shows the way BreastScan was built on purpose to hit its goals. Built with care, it grows when needed, runs without waste, works simply. Strength comes through in every part, setting clear ground for what follows next.

CHAPTER-5 TECHNICAL IMPLEMENTATION & ANALYSIS

5.1 Outline

This part walks through exactly how the BreastScan app was built, followed by a look at how it runs in real situations. What comes next shows the buildup of the system, then shifts into how pieces connect one after another. A close view appears later, revealing behaviour across changing scenarios instead of just ideal ones.

Out of the blueprint comes working code, shaped by steps laid earlier. Coding begins first, while machine learning pieces come together alongside it. OCR features take shape at the same time, built piece by piece. User screens appear next, formed with attention to flow and function. Separate parts live on their own until connections unite them fully. Only then does everything link into one running whole.

From beginning to end, this section walks through the inner workings of the BreastScan app, showing how ideas take shape in real functions. One step at a time, it reveals what happens behind the screen when users interact with the system. Instead of just explaining theory, it ties each idea directly to actual features found inside the software. Through clear examples, it connects design choices to their role in everyday use. Not every detail is obvious up front, yet patterns emerge as the explanation unfolds. With each part laid out plainly, the logic behind decisions becomes visible without extra commentary. What results is a view from within - how code supports purpose.

5.2 Technical Setup and Code Details

Inside Android Studio, that's where BreastScan takes shape. Java handles the behindthe-scenes actions, while XML shapes how things look on screen. One piece at a time, functions live apart but work together. Each part stands alone, yet fits into the whole like a quiet puzzle. Building it this way keeps things clear, step by steady step. Inside the project, several pieces work together - each handling separate jobs like reading what users type, sorting information, pulling text from images, or guessing outcomes through smart algorithms. Starting things off, one central piece opens the door, guiding movement between parts.

Inside the app, screens for typing stuff come from XML files holding boxes, labels, and clickable parts. From there, bits written in Java grab what people type, check if it makes sense, then send it off to the brain of the system.

From within the app, a specific class brings in the TensorFlow Lite model. Inside the assets folder sits the model file, reachable through an Interpreter instance. Prior to reaching the model, raw input shifts into a ByteBuffer structure. Feeding predictions happens only after this transformation completes.

Starting off, Google ML Kit handles the text scanning job. A photo gets snapped or brought in from storage instead of just sitting there. After that comes the reading part - where letters turn into digital words through a special online tool. Words show up as raw data once they're pulled out of the picture. From there, bits of useful info get picked out like finding needles in hay.

A fresh size adjustment happens first, then normalization shapes the image data.
Android's Bitmap tools handle these steps just before sending frames into the model.

A built-in toolkit handles tasks like adjusting value ranges before analysis begins. Because properly shaped inputs lead to better results, each transformation step fits the model's needs.

Challenge	Resolution
Model prediction showing incorrect results	Improved data preprocessing (normalization, scaling) and verified model input format
TensorFlow Lite	Corrected model loading, ByteBuffer input handling,
Challenge	**Resolution**
integration issues	and ensured proper model conversion
OCR extraction inaccuracies	Implemented text filtering and manual verification option for users to edit extracted values
Handling different report formats	Used pattern matching and flexible parsing logic to extract relevant data from varied report layouts
Image prediction inconsistency	Applied image preprocessing techniques like resizing and normalization to improve accuracy
App performance lag on some devices	Optimized code and reduced memory usage for smoother performance on mid-range devices
User input errors in form	Added input validation and error messages to prevent incorrect data entry
UI navigation complexity	Redesigned menu structure for better accessibility and user experience
Data privacy concerns	Implemented on-device processing using TensorFlow Lite to avoid data transmission

Handling large image files	Compressed and resized images before processing to improve speed and efficiency

Table 5.1 Challenges and Resolutions

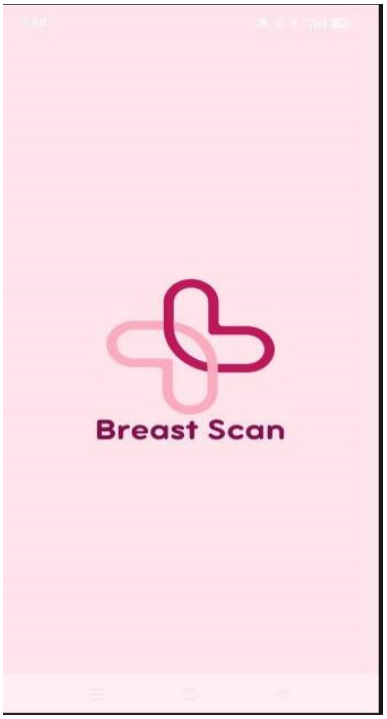
5.3 Prediction Modules Operation

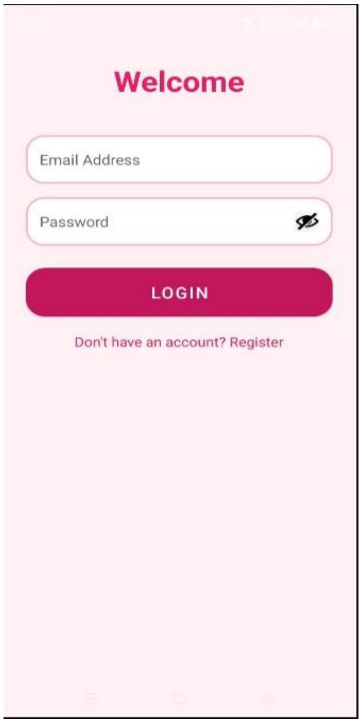
A tool called BreastScan works through three separate parts that guess outcomes in different ways. One part uses forms, another reads text from images, while the third studies pictures directly. Every piece moves step by step, turning inputs into answers without skipping stages.

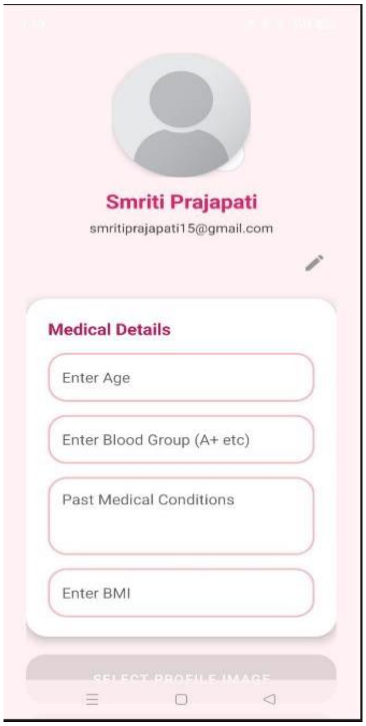
After typing in health details by hand, folks feed info into the prediction tool. Once entered, the software checks each value before adjusting it to fit standard ranges. That cleaned information moves over to a lightweight machine learning setup built with TensorFlow Lite. Out comes a forecast based on what the model learned earlier. Alongside appears a number showing how sure the system feels about its guess. Inside the OCR prediction tool, someone uploads a medical document. From that file, text gets pulled out by optical scanning tech. Relevant numbers pop up through pattern detection. Once found, those figures land inside form fields without manual typing. Before moving forward, people check or change what was captured. After confirmation, info travels toward the learning algorithm for next steps.

A picture shows up when someone adds it. After that a guess happens. The system waits for the file first thing. Once inside, analysis begins right away. Uploading comes before anything else. Then the turn shifts to forecasting. An image arrives; next step follows.

Screenshot of Working Application



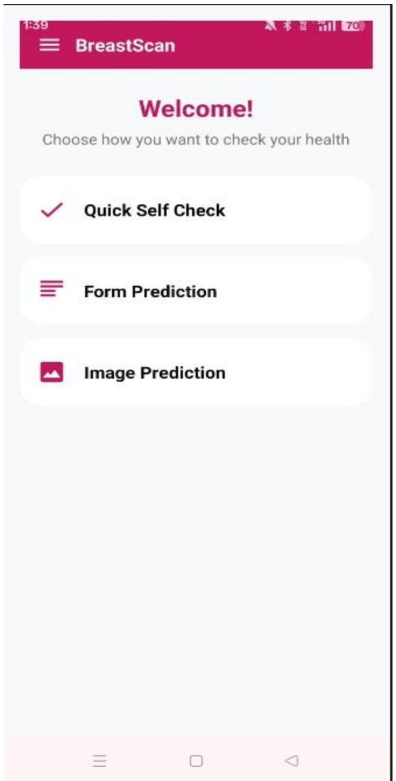


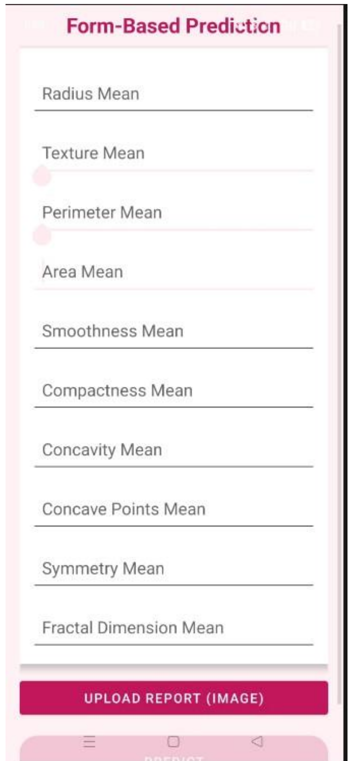


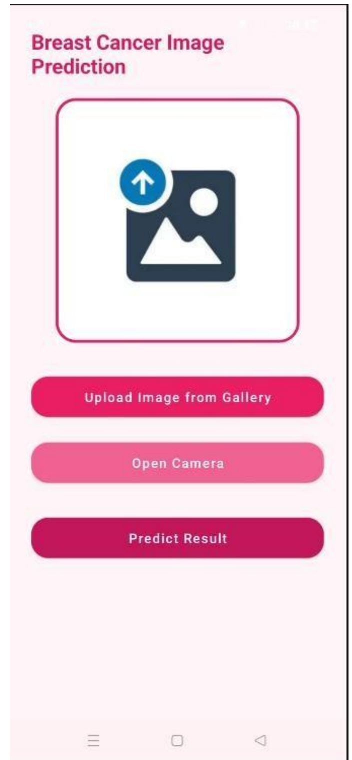
5.1 Splash Screen

5.2 Login Page

5.3 Profile Page







5.4 Welcome Screen 5.5 Form-Based 5.6 Image-Based

5.4 Google ML Kit OCR integration

Inside BreastScan, reading text from images works through Google ML Kit - giving strong support for spotting words in documents. When people share their health records, the system pulls out key details without manual typing.

A single photo of the report kicks things off. After that, the system runs it through ML Kit's text scanner. From there, printed words get spotted and turned into data computers understand.

Figures like readings or scores get pulled out once the text is broken down. Because patterns matter, matches form through coded rules that catch each needed bit. Where digits appear, a system hunts them using preset shapes for clarity. After sorting begins, exact pieces rise - guided by structure - to stand clear.

From there, the data flows into matching spots inside the app. Less handwork needed, fewer mistakes made. Before moving on, people can check the info pulled out - change it if they want.

When text pull fails, the OCR part flags issues quietly. Instead of guessing, it shares what went wrong directly.

A single step forward - OCR fits into the app, lifting how well it works. Efficiency climbs without extra effort.

5.5 Integrate TensorFlow Lite Models

Running quietly inside BreastScan, TensorFlow Lite brings machine learning to life on phones. Instead of draining power, it works smoothly even when resources are tight.

Outside training shapes the model using marked data on breast cancer instances. Once ready, it transforms into TensorFlow Lite form, then lands inside the app's asset collection.

Picture turned into numbers first thing. Once sized right plus made uniform, it feeds the system. Model reads what comes through after prep work finishes.

From the model comes an interpretation shown directly to the user - this becomes what they see as the predicted outcome. Reliability shows up too, not as a guess but measured through a score worked out by the system.

Faster performance comes from running computations directly on the device through TensorFlow Lite. Privacy gets a boost since information stays local instead of traveling online.

5.6 Testing and Validation

Without checking every part carefully, the BreastScan app might miss key details. Because errors can happen anywhere, each feature gets examined in multiple ways.

Sometimes problems show up only under specific conditions, so tests run in varied situations. Accuracy matters most, which means results must match real-world expectations closely. Each method targets a unique piece of how the system behaves overall.

Checking one piece at a time helps spot problems early. When input checks run alone, errors show up clearly. OCR reading gets examined without other steps involved. Predictions from models are tried first in isolation. Every part works right only when tested on its own. Separate trials mean cleaner results overall.

Testing with real people helps see how well the app works in practice. Their thoughts shape changes that make it easier to use.

From start to finish, tests help confirm the app works without hiccups. Smooth performance comes through careful checks along the way. People find it easier to use when each part has been reviewed closely.

5.7 Performance Analysis (Graphs/Charts)

Looking into how well BreastScan works means checking its speed, precision, and system demands. Efficiency shows up in how fast it replies, how correct the results are, while running without heavy load on devices.

When the app runs, how long it takes to reply gets checked so answers come out without waiting. Running on TensorFlow Lite means things move swiftly, cutting down pauses almost entirely.

Fine results come out when the model runs tests on usual data collections. Its success depends strongly on how clean the incoming information appears. Performance checks measure just how close predictions land to real outcomes.

When resources get checked, it helps the app work smoothly on phones. Keeping memory in check, along with steady CPU control, avoids slowdowns.

Look at how numbers behave when shown as pictures instead of raw data. Dots connected by lines often reveal patterns words cannot describe so easily. When a machine answers quickly or slows down, shapes on paper make it visible. Colors spreading across grids point to heavy loads. What seems abstract suddenly fits into real experience through these drawings. Seeing spikes dip or rise tells more than any table ever could.

Performance checks reveal the app runs smoothly while hitting all necessary benchmarks. Efficiency is confirmed through detailed review of its operation under standard conditions.

5.8 Summary

This part walked through how the BreastScan app was built, step by step. From start to finish, it laid out the coding setup alongside key tech choices. Machine learning met optical character reading inside the system. The way pieces fit together showed clarity in design. Every stage unfolded with purpose, showing what runs beneath the surface.

Through the prediction pieces, a path opens - data moves step by step, each part feeding the next. Into that system slips TensorFlow Lite, paired quietly with Google's toolkit for machine learning, both fitting together without force.

This chapter walks through the way BreastScan works, then checks how well it performs. A clear look at setup comes before testing results show what it can do. Through real steps taken, function becomes visible. Performance data follows implementation details closely. Step by step, operation turns into measurable outcomes. What happens during use ties directly to early design choices. Testing reveals effects of each built-in feature. By showing both build and test phases, meaning emerges naturally. Understanding grows from seeing action alongside result.

CHAPTER-6 PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This section walks through what happened when the BreastScan app came together, showing where it fits in everyday medical settings. Its aim sits in checking whether the work actually hit the goals set at the start, also pointing out shifts it brought into health tech spaces.

Early warning begins here - BreastScan helps spot signs of trouble before they grow serious. Built into a phone app, it thinks ahead so people do not have to wait. One part learns from data, another reads medical notes, while the whole runs smoothly on everyday devices. Together, these pieces form something quiet but sharp: a tool that guides without shouting. Not magic, just careful design meeting real need. What stands out here is the way each piece of the system fits together, showing exactly what it can do. Because every part was built with purpose, the overall behavior makes sense when put to work. Testing revealed patterns worth noticing, simply because real performance shaped the outcome. From start to finish, proof comes not from theory but from actual runs under pressure.

The chapter also looks at how BreastScan works outside labs. From clinics to home use, it checks where the tool fits in daily health routines. One place it shows value is doctor offices using it during exams. Another spot people find it useful is tracking changes over time on their own. Real situations shape how it performs, not just theory. What happens in actual use matters most here.

From start to finish, the chapter shows how useful the BreastScan app can be. Its ability to reach more people stands out clearly through real-world use. What matters most is how it opens doors where help was hard to find before.

6.2 Key Features Implementation

From quick checks to detailed insights, BreastScan offers different ways to explore breast health. Instead of just one approach, it combines tools that help recognize warning signs early. Some parts guide you through symptoms, while others explain risk factors clearly. Through practical design, the app supports informed choices without overwhelming. Each element works quietly, making knowledge more reachable day by day.

A prediction tool built around forms is the key feature. Users type in medical details - answers pop up right away. Putting it together meant checking every entry, cleaning the numbers, feeding them into a compact TensorFlow Lite setup. Before anything gets predicted, the system double-checks formatting and correctness behind the scenes.

One key part works by reading text from reports using image scanning tech.

A quick look inside yourself starts here, built to help you notice changes sooner. Simple steps guide each person through checking their own body. It works by making space for real engagement, clear words appear just when needed. Ease matters most - so every screen feels familiar, never confusing. Learning happens quietly, simply, because how things show up makes all the difference.

A profile shows up when someone opens the app. As login steps finish, personal details come into view. Once verified, people move between areas without repeating access checks. Each section links cleanly behind the scenes. Security stays active while switching screens.

What makes it different is how it shares clear facts on breast cancer - like warning signs, why it happens, some ways to lower risk. Learning this gives people useful insight so they can act early when needed.

From start to finish, BreastScan pulls together different tools in one place, blending imaging methods with data analysis so users get clear insights about breast cancer risks without extra steps getting in the way.

6.3 System Outcomes That Matter

From start to finish, building and checking the BreastScan app brought clear outcomes that show how it can help in medical care. What we found came straight from watching how the system worked when put to the test.

Notably, blending several forecasting techniques worked well. Numbers, health records, together with scans feed into the model's output. The scanning part handles pulling words directly from the documents. Key details get spotted without hiccups, then slotted where needed. Less typing by hand comes out of it - tasks move faster now.

Pictures help the tool guess what comes next, adding depth to how things get studied. Though blurry shots can trip it up, performance still climbs when visuals join in. This part sees shapes and patterns others might miss, making sense where words fall short.

Folks who tried it said getting around feels natural, thanks to clear layout choices.

Moving between parts works smoothly because buttons sit where you'd expect them. What stands out is how little time it takes to figure things out. Even first-time clicks land right, since design follows familiar patterns. Fewer steps mean less frustration when completing tasks.

Now here's a different way to stay alert about your body. With just a few taps, people begin noticing changes they might have missed before. Because of these tools, checking in feels less like a chore. When something seems off, folks are more likely to reach out to a doctor. Little by little, habits shift without pressure or fear.

Running smoothly on phones is something it handles well. Thanks to TensorFlow Lite, speed stays high while using minimal power. Without needing the internet, it works anytime, anywhere. Accessibility gets a quiet boost because of this.

From start to finish, findings show the BreastScan app works well at spotting early signs of breast cancer while also helping people pay closer attention. Though quiet in design, it manages to guide users without fuss toward better understanding. Even when used casually, it holds steady in delivering useful insights. Not flashy, yet clear in purpose - its strength lies in simplicity paired with function. In everyday settings, it fits naturally into routines, making check-ins feel routine rather than rare.

6.4 Real-World Applications

From hospitals to home use, BreastScan fits into many everyday health situations. Because it works simply, people find it easy to include in their routines. Whether someone checks symptoms alone or a clinic uses it regularly, the system adapts well. Doctors might rely on it during visits, just as patients might at night after noticing something odd. Even big medical centers can fit it into existing workflows without major changes. How it helps depends on who is using it and when. Not every tool blends this smoothly across such different settings.

BreastScan helps people keep track of their health day by day. Checking how things stand becomes possible, so spotting early warnings happens more often. Since several ways to predict show up, picking one that fits life better makes sense. Sometimes apps help doctors check patients early on. When medical staff look at health details fast, choices get easier. Remote spots without high-tech clinics might rely on these tools more. Where big hospitals are far away, simpler tech steps in. Sometimes apps show up where you least expect them - like helping spread knowledge during health drives. Groups working on outreach might hand out tips using BreastScan, pointing folks toward smarter habits around breast checks. A fast personal check built into the app teams up with clear guides, turning screens into quiet teachers. Not loud, just useful when someone needs to know what comes next. Remote care gets a boost through telemedicine use. Built into digital health systems, it allows doctors to check in from afar. When clinics are hard to reach, this feature makes follow-ups possible. Distance matters less when support travels through screens.

From time to time, this setup shows up in studies focused on information processing. When scientists dig into breast cancer details, they lean on the tool to gather insights - then piece them together for progress.

Few tools adapt as easily to different illnesses, yet this one does - turning a single idea into broad medical use. Its design allows shifts between conditions without losing function, almost like changing lanes on a highway. What begins focused soon spreads, quietly fitting new roles. Flexibility builds in from the start, so hospitals might apply it widely. Not every system handles such jumps, but here the path stays open.

Above all else, BreastScan shows promise where it counts - reaching more people, working faster, yet staying clear about its purpose. Real results prove it can shift how care is shared, step by steady step.

6.5 Inference

One way to tackle health issues shows up in the BreastScan initiative, where smart software meets phone tools. A closer look reveals how pieces like pattern recognition, reading scanned text, and adjusting pictures fit inside one program. What stands out is not just the mix but how each part works together without extra steps. Instead of separate systems, everything runs under one roof. Another thing noticed is how decisions come from data without needing manual checks every time.

Even small details matter when putting such elements side by side. This kind of setup changes what people expect from medical apps on handheld devices.

Testing shows the system delivers precise forecasts without delay. Because processing happens right on the device, speed improves while keeping information private. A straightforward design allows easy access, fitting many different people. Despite varied needs, it stays responsive and clear. Performance remains steady even under changing conditions.

Starting with a quick self-check builds familiarity over time. When people see how things change, they tend to pay closer attention. Learning becomes part of the flow, not something separate. Health shifts feel less distant once small signs are noticed early. Spotting patterns leads some to reach out for help sooner than before. Information fits naturally into daily moments instead of feeling forced. Over weeks, tiny observations add up quietly.

Still, it isn't perfect. Predictions only work as well as the data fed into them, along with how strong the learning algorithm is. Blurry pictures can trip up text scanning tools. Guessing from images still needs more refinement.

Even with its limits, BreastScan helps users get an early look at possible concerns while boosting understanding. What stands out is how it pairs smart software with everyday phones - offering new paths in health care.

Ultimately, success marks the outcome here - goals were hit. Built on solid groundwork, what comes next feels possible. Refinements will come; testing follows. Expansion isn't far off, bringing sharper diagnostics into play. Broader uses in health settings start to take shape once updates settle in.

CHAPTER-7 CONCLUSION

Here ends the story of BreastScan, wrapping up what was done, also pointing toward possible upgrades down the line. What began as an idea now stands tested, its goals either met or adjusted along the way. Through trial and observation, the method showed it can help spot issues earlier than some past approaches. Each result adds weight, yet questions remain that invite further study later on.

One thing led to another when the makers built BreastScan - a phone tool using smart algorithms, text detection from images, plus ways to adjust visuals, all aimed at handling info tied to breast cancer. The best part is the simplicity, so anyone could use it, even if they knew nothing about tech or medicine.

A look at what holds back today's setup begins here, showing spots that call for change. Where things fall short becomes clear when examined closely.

Looking back at how things unfolded while building and checking the system shapes what comes next. Because real experience guided these ideas, they fit more naturally into later efforts. What worked - and what did not - points toward better results down the line.

The integration of intelligent technologies with healthcare creates opportunities for more efficient, accurate, and accessible medical services. This section examines how such tools are not merely used in isolation, but are thoughtfully incorporated into everyday patient care. A tool like BreastScan runs on phones, bringing help where scans are hard to get. Where hospitals lack high-end gear, something basic yet sharp steps in. Early checks become possible without long trips. People learn signs sooner because the app shows what to watch. Waiting too long might drop when alerts come fast. Medical visits could start earlier, simply by tapping a screen. Simple access changes how fast someone acts. Quiet shifts like these reshape who gets seen - and when.

Looking back, this part takes stock of the work done - what it managed to do well, where it fell short, what could come next. Though not perfect, each piece adds context, shaping how the effort fits into broader goals. Some parts worked better than expected, others need more thought later. Progress shows, even if incomplete.
What stands out most is learning through doing.

INDIVIDUAL CONTRIBUTION BY MEMBERS

1. Smriti Prajapati 23BCE11456

My main job was building the core parts of the BreastScan app from the start. From beginning to end, every piece of the system came together through my work on both front and back ends. Smooth navigation and clear layout guided how I shaped the screen designs. Behind the scenes, logic handling, module connections, and information movement fell under my control. Storing details safely while keeping transfer reliable became a daily focus during setup. Putting components into place happened step by step without skipping checks along the way. Each part had to talk correctly - this meant constant testing between sections. Decisions around structure influenced how fast things ran later on. User needs led changes instead of assumptions about what might help. What showed up on devices matched exactly what the code sent behind it. Keeping everything linked properly avoided gaps when pulling records. No outside tools took over tasks better done directly inside the framework. Handling updates required watching interactions down to small shifts. Results stayed consistent because layers were built one at a time. The whole process moved forward only after each section proved stable.

Working alongside model deployment, my role included embedding machine learning systems directly into the app while verifying smooth operation. Testing happened early, often followed by fixes and tuning - each step aimed at stability and speed. Progress moved forward through constant coordination, keeping deadlines in sight across every phase. Final assembly brought pieces together, turning separate parts into one working product.

2. Aditi Raj- 23BAI10178

Aditi here — my role in shaping the project focused on building and refining a machine learning model to predict outcomes from form-based inputs, along with contributing to a breast scan prediction application.

Since the data was not readily available, the process began with collecting and carefully cleaning it to ensure it was suitable for training

. I worked extensively on model training by selecting relevant features and choosing appropriate algorithms to improve prediction accuracy.

Through multiple iterations of training and testing, I fine-tuned the model parameters, which led to more consistent and reliable results.

Careful evaluation at each stage helped strengthen confidence in the model's performance, and over time, predictions aligned more closely with the input data.

As part of deploying the solution, I converted the trained machine learning model into a **pickle (.pkl) file**, making it easier to integrate into the breast scan application for real-time predictions.

This step ensured smooth model loading and efficient performance within the app environment.

In addition to model development, I also contributed to preparing the project presentation, organizing key insights and results in a clear and structured way.

Overall, this work played a significant role in improving the effectiveness of the system while maintaining accuracy and usability.

3. Ghaziah Shoeb- 23BHI10034

Ghaziah here - building the image-driven forecast system inside BreastScan fell to me. Starting off, raw visuals needed sorting, cleaning, before anything else could happen. Instead of skipping steps, each batch got prepped carefully so patterns wouldn't get lost later. Picking how to examine those pictures came next, using tools tuned for spotting subtle differences. Rather than default choices, custom strategies shaped how pixels turned into insights. Algorithms learned from examples, some shallow, others layered deep within networks. Testing followed every round, checking whether guesses matched reality closely enough. Accuracy mattered most - a wrong call wasn't an option when results counted. Each result had to trace back to solid logic, not random luck or vague correlations. In the end, the model stood ready, able to interpret new scans with steady precision. Testing across varied scenarios helped boost how well the model performed. Because of this, the app gained the ability to handle diagnoses using images.

4. Hrithema- 23BCE11243

Hrithema here - my contribution focused on developing the UI of the BreastScan application, along with designing its main logo. The work extended beyond basic visuals, with each design decision aimed at reflecting the purpose and usability of the application. The overall approach was guided by clarity and simplicity, ensuring that the interface remains intuitive and accessible.

The UI was designed with simplicity in mind, enabling users of all backgrounds to easily understand and navigate the application. From layout structure to typography and color selection, each element was carefully chosen to create a seamless interaction between healthcare and technology. Attention to spacing, alignment, and visual hierarchy ensured a clean and user-friendly experience.

In addition, the logo was designed to visually represent the core idea of the application. Every element—from shapes to colors—was thoughtfully selected to convey meaning while maintaining a professional and modern appearance. Together, these efforts helped create a cohesive identity that balances functionality with visual appeal.

5. Tanmay Agnihotri 23BAI10341

Tanmay here — I joined the project at its initial stage and remained actively involved throughout the entire development lifecycle. I contributed to both the technical and organizational aspects of the project, ensuring smooth progress from planning to final delivery.

On the technical side, I supported the development of the application by assisting in structuring the workflow and ensuring proper integration between different components. I helped manage the overall workflow of the project by organizing tasks, tracking progress, and making sure each phase aligned with the planned objectives.

I also contributed to **content creation and documentation**, preparing clear and structured material related to the project, which helped in better understanding and communication of ideas within the team. In addition, I played a key role in **coordinating with the project supervisor**, conveying updates, incorporating feedback, and ensuring that the project stayed aligned with expectations and guidelines.

From a collaboration perspective, I actively facilitated communication among team members, resolved coordination gaps, and maintained a smooth flow of information. I was also involved in **presentation development**, helping design and organize the final project presentation to effectively showcase our work, results, and key insights.

Overall, I contributed significantly by balancing technical support, workflow management, communication, and presentation efforts, ensuring the project was executed efficiently and delivered successfully.

6. Aaditya Jain 23BCE10963

That time working on the project, I joined talks where ideas shaped up slowly. My role was helping keep things moving when the group needed alignment. Thoughts came together best after we traded views at length. Planning steps felt smoother once everyone shared their take. What stood out - listening closely made a difference then.

Working alongside others, I helped keep teamwork steady while keeping messages clear across the group, I contributed during shared tasks.

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BIODATA OF MEMBERS

SMRITI PRAJAPATI Register Number: 23BCE11456 Email: smriti.23bce11456@vitbhopal.ac.in

Right now, Smriti Prajapati studies Computer Science and Engineering at VIT Bhopal University, diving into full-stack development along with building mobile apps. During her time there, she picked up solid skills in Kotlin, using Android Studio, working with Jetpack tools, plus handling coroutines for async tasks. The thing that catches her attention most is App structures that grow smoothly, interfaces that adapt fast, connecting databases without hiccups, then fine-tuning systems for live performance.

Working closely on mobile app design, she shaped clean layouts while guiding how users move from one screen to another. Her efforts cantered around building interface components, organizing system frameworks, along with managing information flow within front-end systems. Technology, in her view, should serve people directly - lifting barriers instead of adding complexity. Over time, she plans to grow into roles combining back-end and mobile engineering, targeting tools that support community needs through smart, inclusive software.

HRITHEMA Register Number: 23BCE11243 Email: hrithema.23bce11243@vitbhopal.ac.in

Starting fresh each time, Hrithema approaches development with clarity and purpose. With a background in computer science, problems are treated as structured challenges, solved through careful thinking and practical implementation. Her work primarily focuses on UI development, where simplicity and usability guide every decision.

Using tools like Android Studio and technologies such as Kotlin, she builds interfaces that are intuitive and responsive. User flows are continuously refined to ensure smooth navigation, while design elements are chosen deliberately to maintain clarity and consistency. Even complex screens are simplified through repeated testing and thoughtful structuring, ensuring accessibility for all users.

Performance and responsiveness remain key priorities. With the use of Coroutines, background tasks are managed efficiently, allowing the application to remain stable and responsive during user interactions. Each component is designed with attention to detail, ensuring that functionality and user experience go hand in hand.

In addition to UI development, she contributed to designing the application’s logo, aligning visual identity with the core purpose of BreastScan. Her role emphasized creating a cohesive experience where design, usability, and performance work together seamlessly.

TANMAY AGNIHOTRI Register Number: 23BAI10341 Email: tanmay.23bai10341@vitbhopal.ac.in

Starting out in computer science, Tanmay Agnihotri studies engineering at VIT Bhopal University with a sharp focus on building apps for web and mobile. Working deeply with Kotlin, he shapes Android applications while weaving together interfaces that respond well to users. Instead of just coding features, his work breathes life into screens through smart layout choices and fluid design logic. Coroutines help him manage background tasks smoothly, keeping apps fast without freezing the display. User experience gains strength under his touch - interactions feel natural, transitions stay clean. Behind every screen lies careful coordination between frontend visuals and server-side functions. His role stitches parts together so data flows quietly where it needs to go. Projects grow more stable because he builds them thinking ahead about how people actually tap and swipe.

What drives Tanmay is how software scales when demand grows, also how data moves without slowing things down. A smooth ride for users matters most, so he shapes interfaces with clean logic, not clutter. Instead of chasing trends, he studies what lasts - frameworks that hold up under pressure, code that adapts quietly. Security isn't an afterthought; it threads into every layer from day one. Over time, he aims to craft tools people rely on daily, especially those serving healthcare, education, civic needs. Systems built right fade into the background, letting purpose lead. That kind of quiet strength defines his path forward.

ADITI RAJ Register Number: 23BAI10178 Email: aditi.23bai10178@vitbhopal.ac.in

Fueled by curiosity, ADITI RAJ dives into Computer Science and Engineering with sharp attention - her path leans heavily toward Machine Learning engineering and fitting models onto devices directly. What stood out was her hand in shaping this effort: building models, trimming their size smartly, then reshaping them to run smoothly on phones. She handles raw data carefully, shapes inputs thoughtfully, trains systems with precision, checks result closely - all using TensorFlow, ScikitLearn, and TensorFlow Lite as go-to helpers. Her work moves quietly but leaves clear marks where performance meets practicality.

Working on this app, she focused hard on a prediction system using Random Forest, making sure it stayed accurate while running smoothly on small devices. Besides tuning numbers, her time went into testing speed and smoothing out how the model works live. What drives her studies sits at the crossroads of fairness in AI, choices shaped by data, and crafting tools that learn smart but act right. Thinking ahead, she aims to build models people can count on - especially where lives depend on them, like medicine or emergency services.

GHAZIAH SHOEB Register Number: 23BHI10034 Email: ghaziah.23bhi10034@vitbhopal.ac.in

Shoeing a clear path through tech challenges, GHAZIAH SHOEB studies Computer Science while diving deep into cloud setups, server-side coding, besides data storage systems. When the team needed solid groundwork, she stepped in - handling core pieces of the backend by leaning on Supabase paired with PostgreSQL. Authentication logic and database blueprints are designed by her. Even sensitive API routes got her careful touch, along with locks built on JWT methods. Working behind the scenes, her contributions held structure together without drawing attention. Precision runs through each layer she touches, especially where security meets functionality. Not loud, yet vital - the kind of presence that keeps digital frameworks standing.

Security stayed tight behind the scenes, scalability grew without strain, efficiency held steady - data never slipped loose during any phase. Starting fresh each time, Ghaziah leans into crafting cloud setups built for instant data flow plus rock-solid uptime. Far ahead lies her aim: shaping protected digital backbones, especially where health records live and system trust can't waver.

AADITYA JAIN Register Number: 23BCE10963 Email: aaditya.23bce10963@vitbhopal.ac.in

A young coder named AADITYA JAIN studies Computer Science and Engineering at VIT Bhopal University, slowly leaning into software creation and new digital tools. While learning, he dives into frontend design alongside backend logic, cloud setups, also database structures - each piece shaping his core skills. His time in college hasn't just been about code; it's been shaped by grasping how real users interact with apps, how information moves without hiccups, even how systems grow without breaking. Though still early, his path centres on making tech that works clearly, runs smoothly, stands firmly.

Right now, Aaditya builds things using Kotlin and Android Studio while diving into web tools at the same time. Through steady effort, he grows stronger in full-stack work step by step. Security matters too - so databases with tight protection catch his attention often. Connecting systems via APIs comes up regularly in his routine lately. Real-time flows keep showing up as a key piece of what he studies. Each skill fits where it should when new projects take shape ahead.

One thing he holds close is always learning new things, while sharpening how he tackles problems along with deepening his grasp of tech details. His motivations are

- future where software works well, stands strong under pressure, yet brings fresh solutions to everyday challenges - part of a broader effort that values ethics just as much as progress.

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MADHYA PRADESH – 466114

BONAFIDE CERTIFICATE

Certified that this project report titled **"BreastScan: An AI- Powered Mobile Application for Breast Cancer Prediction"** is the Bonafide work of **"Smriti Prajapati** (23BCE11456); **Hrithema** (23BCE11243); **Aditi Raj** (23BAI10178); **Tanmay Agnihotri** (23BAI10341); **Ghaziah Shoeb** (23BHI10034); **Aaditya Jain** (23BCE110963)**"** who carried out the project work under my supervision. Certified further that to the best of my knowledge, the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

PROJECT GUIDE
Dr. Praveen Lalwani
Assistant Professor
School of Computer Science and Engineering
VIT BHOPAL UNIVERSITY

27/03/26
REVIEWER 2

Dr. Ankit Pal
Assistant Professor
School of Applied Science and Language
VIT Bhopal University

REVIEWER 1

Dr. Buvaneswari P. R.
Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal University

The EPICS Phase 2 is held on 30th March, 2026

VIT BHOPAL UNIVERSITY, KOTHRI KALAN, SEHORE
MADHYA PRADESH – 466114

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PROJECT GUIDE
Dr. Praveen Lalwani
Assistant Professor
School of Computer Science and Engineering
VIT BHOPAL UNIVERSITY

27/03/26

REVIEWER 1
Dr. Buvaneswari P. R.
Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal University

REVIEWER 2
Dr. Ankit Pal
Assistant Professor
School of Applied Science and Language
VIT Bhopal University

The EPICS Phase 2 is held on ____

Declaration of Originality

We hereby declare that this report entitled **"BreastScan: An AI-Powered Mobile Application for Breast Cancer Prediction"** represents our original work carried out for the EPICS project as students of **VIT Bhopal University**. To the best of our knowledge, this report contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution.

The research, design, implementation, and documentation presented in this report are based on our own efforts and understanding developed during the course of this project. All external sources of information, including research papers, articles, datasets, books, websites, frameworks, and tools, have been properly acknowledged in the References section.

We take full responsibility for the authenticity of the work and confirm that it adheres to the academic integrity guidelines prescribed by VIT Bhopal University.

Date

Reg No & Name

23BCE11456- SMRITI PRAJAPATI Smriti Prajapati

23BCE11243- HRITHEMA Hrithema

23BAI10178- ADITI RAJ A. Raj

23BAI10341- TANMAY AGNIHOTRI Tanmay Agnihotri

23BHI10034- GHAZIAH SHOEB GS

23BCE10963- AADITYA JAIN Aaditya